ravel

Grafhen
The first noisefree fully homomorphic cryptosystem

Ravel Technologies

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1. The problem

Innovation and growth in the digital economy will inevitably stagnate without stringent data privacy and security. Indeed, no company wants to risk their sensitive data exposed, nor do individuals.

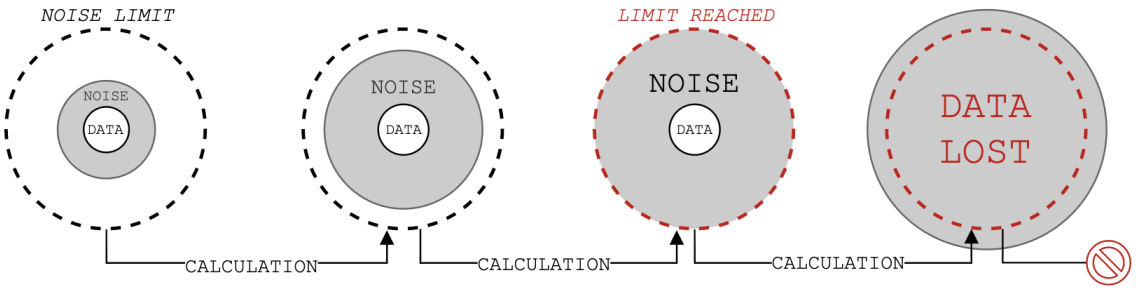
This growth, which feeds in large part on all kinds of data, is in fact hampered because it is synonymous with loss of control over their data for individuals, companies and even states. Coincidentally, unencrypted data processing is a boon for hackers; as a result cyberthreats keep increasing sharply.

Enclave-based confidential computing has so far been merely used as a fragile substitute for real cryptographic protections. Indisputable security and privacy guarantees can only be built on scientifically recognized cryptography rather than historically unreliable and opaque hardware regularly patched.

Enter FHE, the technology that allows to process encrypted data without decrypting it. It might seem surprising at first, but the data indeed stays encrypted while computations are run. This new privacy standard is the promise of a new era of digital acceleration.

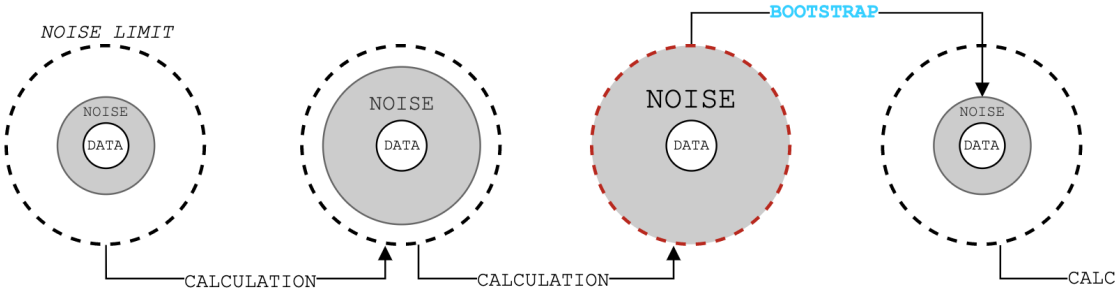
2. Why is traditional FHE failing to scale?

The quest to design an FHE scheme allowing both addition and multiplication remained an open problem for 30 years, until Gentry's seminal construction in 2009 defined the first Fully Homomorphic Encryption (FHE) scheme. **Traditional FHE schemes require noise** to be added to the ciphertext; otherwise, it would be trivial to recover the plaintext. When computing on encrypted data, the noise grows and if nothing was done, at some point the plaintext could no longer be recovered.



Noise growth in traditional FHE

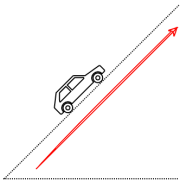
To be able to run more computations, **to keep the noise under control, a "bootstrap" operation is needed. It successfully reduces the noise in the ciphertext but is very costly**. A bootstrap is typically needed each time a multiplication is computed.



Bootstrap

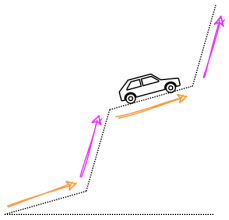
Tremendous efforts have been devoted to accelerating this bootstrap, but these require increasingly expensive hardware, and performance remains on the order of 10 million times slower than computation on cleartext. This cost radically hinders practical applications.

To illustrate with a metaphor, noise management in homomorphic encryption is like a car driving uphill. Moving forwards (to the right) requires also spending energy to fight gravity; likewise, to produce meaningful computations, the server must work hard to perform bootstraps.



FHE w/ bootstrap

Another approach, leveled homomorphic encryption (e.g. CKKS is in this category), works like prepaid cards. It allows a fixed number of multiplications before either producing a result or requiring an even more expensive bootstrap to "top up" the noise budget. In other words, it is useful only for short sequences of operations (typically around 5), which severly limits applications. Returning to our metaphor, the car rides a gentler slope for a while but periodically needs to climb a steeper one than in the previous case. The average slope is roughly the same.



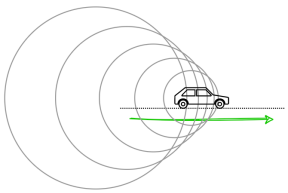
Leveled FHE

In summary, noise has been inherent to FHE for 16 years and has been a hard barrier to efficient fully homomorphic computation.

3. GRAFHEN – breaking the noise barrier

To free FHE from the burden of noise, Ravel decided to try a radically different path, breaking from the lineage of FHE schemes derived from Gentry's work on bootstrapping and noise management, and developed a new FHE scheme: GRAFHEN (GRoup-based Fully Homomorphic Encryption without Noise).

By relying on an innovative and carefully engineered architecture based on groups rather than lattices, **GRAFHEN is the first cryptography scheme to offer FHE without adding noise to the encrypted data and thus without the need for the costly bootstrapping operation**. Sustaining the metaphor, it is as if our car could finally cruise at supersonic speed on a flat highway, without any painful, slowing climbs.

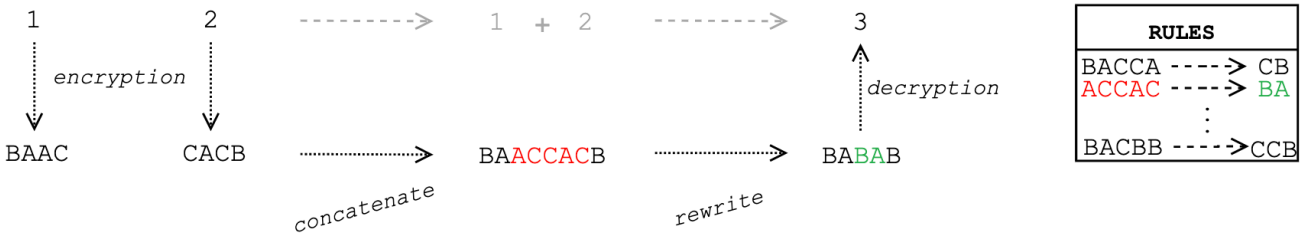


Noisefree FHE

A key point of how GRAFHEN works is that encrypted values are not numeric, as in traditional encryption schemes, but symbolic, namely letters and words. Therefore intensive polynomial operations are replaced by concatenation of words, then search-and-replace to shorten them.

GRAFHEN therefore allows operations – from the simplest ones like addition and multiplication to the most complex ones such as gradient descent or matrix multiplications widely used in AI – to be performed on encrypted data considerably more efficiently than in traditional FHE schemes.

Here is a glimpse at how an addition plays out:

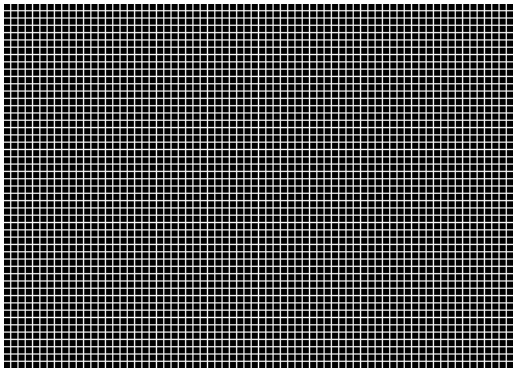


The numbers 1 and 2 are encrypted as words (shorter in this example than in reality). To compute the sum, the operation on ciphertexts consists in:

1. concatenating both words
2. reducing the result using rewrite rules: the rule on the second line says "replace each occurrence of `ACCAC` by `BA`)".

This rewrite operation allows to keep stable the length of ciphertexts throughout computations.

Finally, to put the magnitude of the breakthrough in perspective, **GRAFHEN V0 is already 3,500 times faster than traditional FHE** (as benchmarked against OpenFHE).



Traditional FHE time

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GRAFHEN time